

LA HW

Included exercises: 1, 3, 5, 7, 9, 13, 15, 17, 21, 25, 27, 29, 33, 35, 39, 41, 53, 55, 57, 59, 61, 65, 69, 71

Exercises 1-12

In Exercises 1–12, a matrix and its characteristic polynomial are given. Find the eigenvalues of each matrix and determine a basis for each eigenspace.

Original Exercise 1

$$\begin{bmatrix} 3 & -3 \\ 2 & 8 \end{bmatrix}, \quad (t - 5)(t - 6)$$

Original Exercise 3

$$\begin{bmatrix} -10 & 6 \\ -15 & 9 \end{bmatrix}, \quad t(t + 1)$$

Original Exercise 5

$$\begin{bmatrix} 6 & -5 & -4 \\ 5 & -3 & -5 \\ 4 & -5 & -2 \end{bmatrix}, \quad -(t + 3)(t - 2)^2$$

Original Exercise 7

$$\begin{bmatrix} 6 & -4 & -4 \\ -8 & 2 & 4 \\ 8 & -4 & -6 \end{bmatrix}, \quad -(t - 6)(t + 2)^2$$

Original Exercise 9

$$\begin{bmatrix} 0 & 2 & 1 \\ 1 & -1 & -1 \\ 4 & 4 & -3 \end{bmatrix}, \quad -(t+3)(t+2)(t-1)$$

Exercises 13-24

In Exercises 13–24, find the eigenvalues of each matrix and determine a basis for each eigenspace.

Exercise 13: $\begin{bmatrix} 1 & 3 \\ 0 & -4 \end{bmatrix}$ Exercise 14: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Original Exercise 13

$$\begin{bmatrix} 1 & 3 \\ 0 & -4 \end{bmatrix}$$

Original Exercise 15

Exercise 15: $\begin{bmatrix} -3 & -4 \\ 12 & 11 \end{bmatrix}$

$$\begin{bmatrix} -3 & -4 \\ 12 & 11 \end{bmatrix}$$

Original Exercise 17

$$\begin{bmatrix} -7 & 5 & 4 \\ 0 & -3 & 0 \\ -8 & 9 & 5 \end{bmatrix}$$

Original Exercise 21

$$\begin{bmatrix} -4 & 0 & 2 \\ 2 & 4 & -8 \\ 2 & 0 & -4 \end{bmatrix}$$

Original Exercise 25

Find a basis for each eigenspace.

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -x_1 + 6x_2 \\ -8x_1 + 13x_2 \end{bmatrix}, \quad (t-5)(t-7)$$

Original Exercise 27

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -10x_1 - 24x_2 \\ \dots \end{bmatrix}, \quad (t-6)(t-2)$$

Original Exercise 29

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} -2x_2 + 4x_3 \\ -3x_1 + x_2 + 3x_3 \\ -x_1 + x_2 + 5x_3 \end{bmatrix},$$

$$-(t+2)(t-4)^2$$

Original Exercise 33

determine a basis for each eigenspace.

$$T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} -4x_1 + x_2 \\ -2x_1 - x_2 \end{bmatrix}$$

Original Exercise 35

$$T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} 2x_2 \\ -10x_1 + 9x_2 \end{bmatrix}$$

Original Exercise 39

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} -3x_1 \\ -8x_1 + x_2 \\ -12x_1 + x_3 \end{bmatrix}$$

Original Exercise 41

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \\ 3x_1 + 2x_2 + 5x_3 \\ 4x_1 + 5x_2 + 6x_3 \end{bmatrix}$$

Show that $\begin{bmatrix} 6 & -7 \\ 4 & -3 \end{bmatrix}$ has no real eigenvalues.

Exercises 53-72

In Exercises 53–72, determine whether the statements are true or false.

Original Exercise 53

ments are true or false.

If two matrices have the same characteristic polynomial, then they have the same eigenvectors.

Original Exercise 55

The characteristic polynomial of an $n \times n$ matrix is a

Original Exercise 57

The eigenvectors of a matrix are equal to those of its

Original Exercise 59

Every $n \times n$ matrix has an eigenvector in $\mathcal{R}^n$.

Original Exercise 61

The characteristic polynomial of an $n \times n$ matrix can be written $c(t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n)$ for some real num-

Original Exercise 65

If λ is an eigenvalue of multiplicity 1 for a matrix A , then the dimension of the eigenspace of A corresponding to λ

Original Exercise 69

The scalar 1 is an eigenvalue of I_n .

Original Exercise 71

A square zero matrix has no eigenvalues.